Скачивание данных

Yersinia pestis - SRR10259779

1. скачивание данных секвенирования из базы SRA

```
prefetch SRR10259779
```

2. конвертация в FASTQ

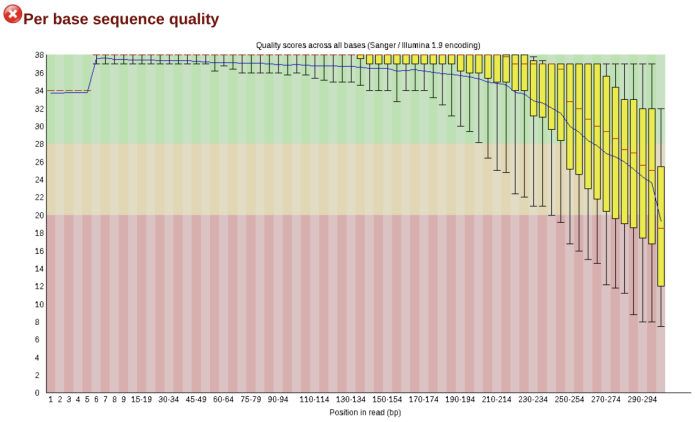
```
fasterq-dump SRR10259779/

spots read      : 494,351
reads read      : 988,702
reads written   : 988,702
```

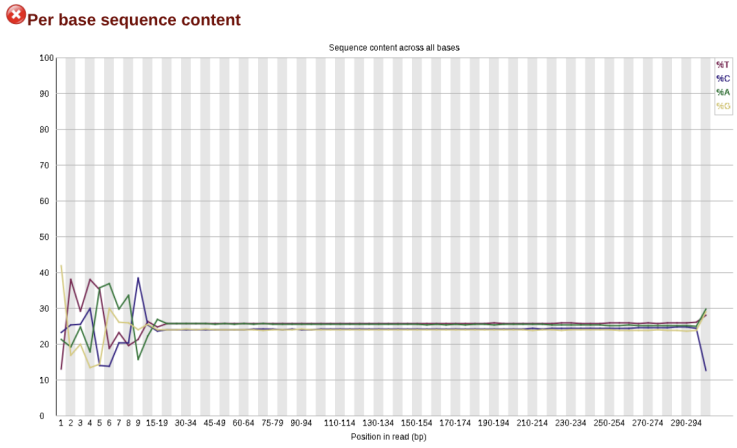
Контроль качества (QC) и фильтрация

Первичный QC

оценка качества сырых данных

```
fastqc SRR10259779_1.fastq SRR10259779_2.fastq
```





Фильтрация fastp

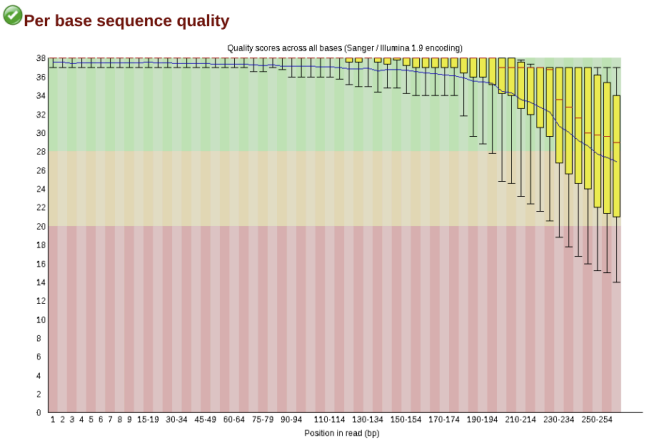
```
fastp -i SRR10259779_1.fastq -I SRR10259779_2.fastq -o SRR10259779_1.filtered.fastq -O
SRR10259779_2.filtered.fastq -q 20 --trim_front1 20 --trim_front2 20 --max_len1 260 --max_len2 260
```

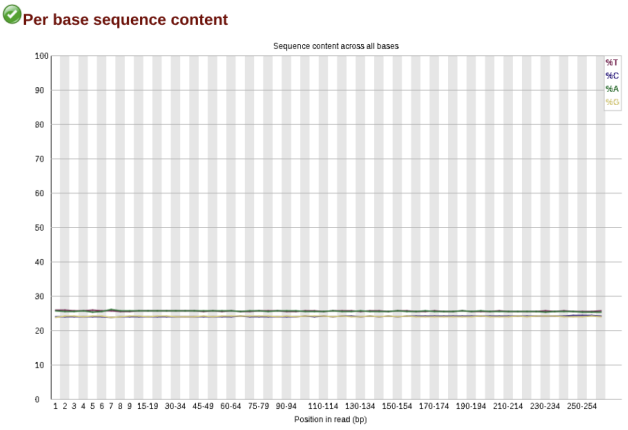
Параметры:

- `-q 20` — обрезка концов с качеством ниже 20
- `--trim_front1/2 20` — удаление первых 20 нуклеотидов
- `--max_len1/2 260` — обрезка чтений до 260 п.н.

Финальный QC

```
fastqc SRR10259779_1.filtered.fastq SRR10259779_2.filtered.fastq
```





Вывод: После фильтрации качество выровнялось, GC-состав стабилизировался, адаптеры удалены.

Сборка генома (SPAdes)

сборка генома на очищенных данных

```
spades.py -1 SRR10259779_1.filtered.fastq -2 SRR10259779_2.filtered.fastq -o output_spades/
```

```
ls output_spades/
```

```
assembly_graph_after_simplification.gfa  contigs.fasta  input_dataset.yaml  K55   params.txt       scaffolds.fasta     tmp
assembly_graph.fastg                     contigs.paths  K127                K77   pipeline_state   scaffolds.fasta.fai
assembly_graph_with_scaffolds.gfa        corrected      K21                 K99   run_spades.sh    scaffolds.paths
before_rr.fasta                          dataset.info   K33                 misc  run_spades.yaml  spades.log
```

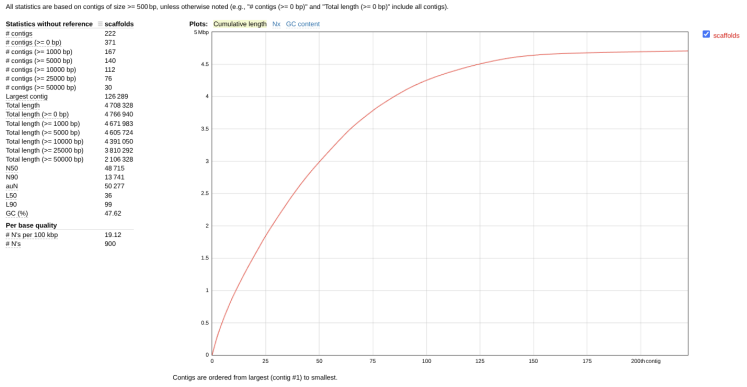
Результат:

Сборка завершена, получены скаффолды.

Оценка качества сборки (QUAST)

```
quast.py --output-dir quast_output output_spades/scaffolds.fasta
```

`report.html`



All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Statistics without reference	scaffolds
# contigs	222
# contigs (>= 0 bp)	371
# contigs (>= 1000 bp)	167
# contigs (>= 5000 bp)	140
# contigs (>= 10000 bp)	112
# contigs (>= 25000 bp)	76
# contigs (>= 50000 bp)	30
Largest contig	126 289
Total length	4 708 328
Total length (>= 0 bp)	4 766 940
Total length (>= 1000 bp)	4 671 983
Total length (>= 5000 bp)	4 605 724
Total length (>= 10000 bp)	4 391 050
Total length (>= 25000 bp)	3 810 292
Total length (>= 50000 bp)	2 106 328
N50	48 715
N90	13 741
auN	50 277
L50	36
L90	99
GC (%)	47.62
Per base quality	
# N's per 100 kbp	19.12
# N's	900

Вывод:

Сборка фрагментирована (N50 48 Kb), но биологически полная (GC% верный, длина близка к референсу).

Оценка полноты генов (BUSCO)

```
busco -i output_spades/scaffolds.fasta -l enterbacterales_odb10 --offline --download_path
busco_asm/busco_downloads -m genome -o busco_output -f
```

```
---------------------------------------------------
|Results from dataset enterobacterales_odb10       |
---------------------------------------------------
|C:99.5%[S:99.3%,D:0.2%],F:0.5%,M:0.0%,n:440       |
|438    Complete BUSCOs (C)                        |
|437    Complete and single-copy BUSCOs (S)        |
|1      Complete and duplicated BUSCOs (D)         |
|2      Fragmented BUSCOs (F)                      |
|0      Missing BUSCOs (M)                         |
|440    Total BUSCO groups searched                |
---------------------------------------------------
```

`short_summary.txt`

```
cat busco_output/run_enterobacterales_odb10/short_summary.txt
```

```
        ***** Results: *****

        C:99.5%[S:99.3%,D:0.2%],F:0.5%,M:0.0%,n:440
        438     Complete BUSCOs (C)
        437     Complete and single-copy BUSCOs (S)
        1       Complete and duplicated BUSCOs (D)
        2       Fragmented BUSCOs (F)
        0       Missing BUSCOs (M)
        440     Total BUSCO groups searched

Assembly Statistics:
        371     Number of scaffolds
        380     Number of contigs
        4766940 Total length
        0.019%  Percent gaps
        48 KB   Scaffold N50
        45 KB   Contigs N50


Dependencies and versions:
        hmmsearch: 3.4
        bbmap: 39.06
        prodigal: 2.6.3
        busco: 5.5.0
```

`full_table.tsv`

```
cat busco_output/run_enterobacterales_odb10/full_table.tsv
```

```
# The lineage dataset is: enterobacterales_odb10 (Creation date: 2024-01-08, number of genomes: 212, number of BUSCOs: 440)
# Busco id      Status  Sequence        Gene Start      Gene End        Strand  Score   Length  OrthoDB url     Description
312at91347      Complete        NODE_97_length_14966_cov_20.775457      4493    8521    -       2307.5  1146    https://v10-1.orthodb.org/?query=312at91347 DNA-directed RNA polymerase, subunit 2, hybrid-binding domain
373at91347      Complete        NODE_97_length_14966_cov_20.775457      144     4364    -       2404.5  1205    https://v10-1.orthodb.org/?query=373at91347 DNA-directed RNA polymerase subunit beta
854at91347      Complete        NODE_16_length_62219_cov_20.196789      9498    12395   +       1410.3  813     https://v10-1.orthodb.org/?query=854at91347 Valine--tRNA ligase
1674at91347     Complete        NODE_11_length_67997_cov_19.065449      22338   24920   +       1213.8  709     https://v10-1.orthodb.org/?query=1674at91347        Leucine--tRNA ligase
2121at91347     Complete        NODE_2_length_120188_cov_19.188830      19678   22299   +       1299.8  712     https://v10-1.orthodb.org/?query=2121at91347        DNA topoisomerase I, type IA
2325at91347     Complete        NODE_14_length_64556_cov_18.647628      43078   46740   +       1318.2  950     https://v10-1.orthodb.org/?query=2325at91347        UvrD-like DNA helicase, C-terminal
2649at91347     Complete        NODE_47_length_39972_cov_20.352707      22890   26405   +       1609.4  959     https://v10-1.orthodb.org/?query=2649at91347        DNA-directed DNA polymerase
2659at91347     Complete        NODE_86_length_19650_cov_19.567354      3491    6154    -       1508.8  775     https://v10-1.orthodb.org/?query=2659at91347        Pyruvate dehydrogenase E1 component
3839at91347     Complete        NODE_24_length_57898_cov_19.742812      20812   23556   +       1308.8  747     https://v10-1.orthodb.org/?query=3839at91347        NADH-quinone oxidoreductase
3884at91347     Complete        NODE_117_length_3425_cov_18.728529      3239    6472    -       1754.5  932     https://v10-1.orthodb.org/?query=3884at91347        Carbamoyl-phosphate synthase large chain
3734at91347     Complete        NODE_71_length_26287_cov_25.747095      22493   25207   +       1421.9  786     https://v10-1.orthodb.org/?query=3734at91347        SecA, preprotein cross-linking domain
4210at91347     Complete        NODE_9_length_77670_cov_18.772913       22959   25313   +       1306.0  682     https://v10-1.orthodb.org/?query=4210at91347        Lon protease
4269at91347     Complete        NODE_81_length_21541_cov_19.216414      11647   13845   -       927.1   688     https://v10-1.orthodb.org/?query=4269at91347        Primosomal protein N'
4411at91347     Complete        NODE_17_length_61568_cov_19.635537      10054   12729   -       1355.6  740     https://v10-1.orthodb.org/?query=4411at91347        DNA gyrase subunit A
4459at91347     Complete        NODE_44_length_16529_cov_18.473584      5891    8143    -       884.8   610     https://v10-1.orthodb.org/?query=4459at91347        Organic solvent tolerance-like, N-terminal
4673at91347     Complete        NODE_35_length_48930_cov_19.303875      180     2207    -       996.9   551     https://v10-1.orthodb.org/?query=4673at91347        Methionine--tRNA ligase
4867at91347     Complete        NODE_17_length_61568_cov_19.635537      14213   16581   +       1326.3  660     https://v10-1.orthodb.org/?query=4867at91347        Ribonucleoside-diphosphate reductase
5022at91347     Complete        NODE_26_length_53078_cov_19.536902      49954   52368   +       1218.2  606     https://v10-1.orthodb.org/?query=5022at91347        DNA gyrase subunit B
5862at91347     Complete        NODE_78_length_26668_cov_18.557665      21528   24344   +       1282.8  779     https://v10-1.orthodb.org/?query=5862at91347        Isoleucine--tRNA ligase
5872at91347     Complete        NODE_98_length_10105_cov_19.773175      3762    6569    -       1285.2  732     https://v10-1.orthodb.org/?query=5872at91347        2-oxoglutarate dehydrogenase E1 component
```

```
grep "Duplicated" busco_output/run_enterobacterales_odb10/full_table.tsv | column -t

80874at91347  Duplicated  NODE_74_length_25656_cov_20.940421  8204  8596  +  243.7  129  https://v10-
1.orthodb.org/?query=80874at91347  Ribosomal  protein  S14

80874at91347  Duplicated  NODE_74_length_25656_cov_20.940421  7865  8170  +  221.7  100  https://v10-
1.orthodb.org/?query=80874at91347  Ribosomal  protein  S14
```

```
grep "Fragmented" busco_output/run_enterobacterales_odb10/full_table.tsv | column -t

29797at91347  Fragmented  NODE_165_length_1144_cov_20.778761  262    1092   +  338.2  221  https://v10-
1.orthodb.org/?query=29797at91347  Chorismate   mutase

73655at91347  Fragmented  NODE_30_length_51129_cov_19.287793  38125  38475  -  202.0  105  https://v10-
1.orthodb.org/?query=73655at91347  Translation  initiation  factor  3,  C-terminal
```

- Complete: **99.5%**
- Duplicated: 0.2% (1 ген)
- Fragmented: 0.5% (2 гена)
- Missing: 0%

Проблемные гены:

- **Duplicated**: Ribosomal protein S14 (NODE_74, две копии)
- **Fragmented**: Chorismate mutase (NODE_165), Translation initiation factor 3 (NODE_30)

Вывод:
Сборка содержит весь генетический материал (0% пропусков), несмотря на фрагментированность.

Скэффолдинг с референсом (RagTag)

скачиваем [NCBI RefSeq assembly GCF_000009065.1]

Download datasets API FTP

		Actions
NCBI RefSeq assembly	GCF_000009065.1	⋮
Submitted GenBank assembly	GCA_000009065.1	⋮
Taxon	Yersinia pestis CO92	
Strain	CO92	
Submitter	Sanger Institute	
Date	Oct 15, 2001	

коррекция

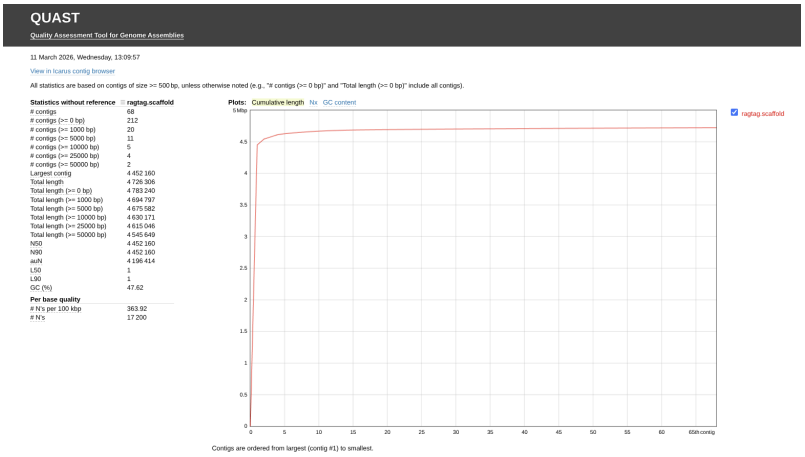
```
ragtag.py correct GCF_000009065.1_ASM906v1_genomic.fna output_spades/scaffolds.fasta -t 8 -o ragtag_correct/
```

скэффолдинг

```
ragtag.py scaffold -t 8 GCF_000009065.1_ASM906v1_genomic.fna ragtag_correct/ragtag.correct.fasta -t 8 -o
ragtag_scaffold
```

```
quast.py --output-dir ragtag_output ragtag_scaffold/ragtag.scaffold.fasta
```

`report.html`



QUAST
Quality Assessment Tool for Genome Assemblies

11 March 2026, Wednesday, 13:09:57
View in Icarus contig browser
All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Statistics without reference	ragtag.scaffold
# contigs	68
# contigs (>= 0 bp)	212
# contigs (>= 1000 bp)	20
# contigs (>= 5000 bp)	11
# contigs (>= 10000 bp)	5
# contigs (>= 25000 bp)	4
# contigs (>= 50000 bp)	2
Largest contig	4 452 160
Total length	4 726 306
Total length (>= 0 bp)	4 783 240
Total length (>= 1000 bp)	4 694 797
Total length (>= 5000 bp)	4 675 582
Total length (>= 10000 bp)	4 630 171
Total length (>= 25000 bp)	4 615 046
Total length (>= 50000 bp)	4 545 649
N50	4 452 160
N90	4 452 160
auN	4 196 414
L50	1
L90	1
GC (%)	47.62
Per base quality	
# N's per 100 kbp	363.92
# N's	17 200

Plots: Cumulative length Nx GC content
ragtag.scaffold
Contigs are ordered from largest (contig #1) to smallest.

Вывод

- Количество контигов сократилось с 222 до 68
- N50 вырос с 48.7 Kb до 4.45 Mb (улучшение в 91 раз!)
- Половина генома собрана в один контиг (L50 = 1)
- Добавлено 17,200 пропусков (N) для соединения контигов, что составляет 0.36% генома

скэффолдинг улучшил целостность сборки, приблизив её к хромосомному уровню.